

A Schur Bridge from Nested Clustered Optimization to Global Minimum Variance

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Abstract

Nested clustered optimization allocates within each cluster from the cluster’s own covariance block and then across the resulting cluster portfolios. Block inversion says the unconstrained minimum-variance portfolio has the same two-tier shape, with each block replaced by its Schur complement against every other asset. Conditioning instead on one knot from each other cluster truncates the conditioning set in the manner of a Vecchia approximation, and we give the rank-one model of cross-cluster dependence under which it is exact. Damping the complement by $\gamma \in [0, 1]$ then gives a bridge with nested clustered optimization at $\gamma = 0$ and the global optimum at $\gamma = 1$, with no linear solve larger than a cluster or the number of clusters. Under estimation error the optimal γ can be strictly interior and full coupling can remain locally optimal, and we give the local theorem at the minimum-variance end with exact examples of both.

1 Introduction

Two-stage cluster methods partition the assets, allocate inside each cluster, and then allocate across clusters. In nested clustered optimization, NCO [1], the inner step optimizes each cluster on its own covariance block and the outer step optimizes across the resulting cluster portfolios. The outer problem has one dimension per cluster, which is the source of the method’s numerical stability.

The inner step reads only the cluster’s own block. The only channel by which one cluster’s weights respond to another cluster is its scalar budget.

Schur-complementary allocation [2] removes the same blind spot from hierarchical risk parity [3]. At each split of a bisection tree it replaces a block by its Schur complement against the sibling block, damped by $\gamma \in [0, 1]$, and carries a companion vector alongside. The result is a bridge with hierarchical risk parity at $\gamma = 0$ and minimum variance at $\gamma = 1$.

This note builds the corresponding bridge for a flat partition. The left end is NCO and the right end is the unconstrained global minimum-variance portfolio. The obstacle is cost, since the complement of a cluster against everything else needs an inverse of nearly full size.

Spatial statistics meets the same obstacle by conditioning on a few chosen indices rather than on everything [4]. The conditionals there are Schur complements, so the device transfers directly.

We borrow its vocabulary. One member of each cluster is its *knot*, as in the low-rank spatial models of Banerjee et al. [5] that Katzfuss and Guinness [6] place inside the Vecchia framework. The other clusters’ knots are a cluster’s *conditioning set*. We state the model under which conditioning on that set loses nothing.

Section 6 asks where to sit on the bridge when the covariance is estimated. Part of the analysis of the tree in [7] transfers, and exact examples show that both an interior optimum and a locally optimal full coupling occur.

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All exactness statements concern the unconstrained, fully invested, long-short problem. With box or long-only constraints no block decomposition of the optimum is exact, here or on the tree.

2 Setup

Let $\Sigma \succ 0$ be the covariance of n assets partitioned into clusters $I_1, \dots, I_k$. Cluster i has a knot $p_i \in I_i$ and remaining members $J_i = I_i \setminus \{p_i\}$. Write $P = \{p_1, \dots, p_k\}$ and $P_{-i} = P \setminus \{p_i\}$. For index sets X, Y write Σ_{XY} for the corresponding block, Σ_{ii} for the block of cluster i , and $-i$ for the complement of I_i .

A two-stage method chooses a portfolio v_i for each cluster, supported on I_i with $\mathbf{1}^\top v_i = 1$, and then budgets across the clusters. With V the $n \times k$ matrix of cluster portfolios,

$$a = \text{alloc}(V^\top \Sigma V), \quad w = Va. \quad (1)$$

NCO takes v_i to be the minimum-variance portfolio of Σ_{ii} and minimum variance for the outer allocation.

3 Block inversion

Fix a vector u and a cluster i . Block inversion of Σ on the partition $(I_i, -i)$ gives [2, App. A]

$$(\Sigma^{-1}u)_{I_i} = (\Sigma_{ii}^c)^{-1}b_i, \quad (2)$$

where

$$\Sigma_{ii}^c = \Sigma_{ii} - \Sigma_{i,-i} \Sigma_{-i,-i}^{-1} \Sigma_{-i,i}, \quad b_i = u_{I_i} - \Sigma_{i,-i} \Sigma_{-i,-i}^{-1} u_{-i}. \quad (3)$$

For a positive-definite Q and a vector b we call $Q^{-1}b$ a *generalized minimum-variance direction*. Up to scale it minimizes $x^\top Qx$ subject to $b^\top x = 1$, which is ordinary minimum variance only when $b = \mathbf{1}$.

With $u = \mathbf{1}$, the blocks of the global minimum-variance direction $\Sigma^{-1}\mathbf{1}$ are therefore generalized minimum-variance directions, one per cluster, on the cluster's covariance conditional on everything else. There is no cross-cluster optimizer in the global optimum. The cluster totals are whatever the stacked directions sum to, and every piece of cross-cluster information enters through the conditioning.

NCO has the same shape with the conditioning removed. Restoring it costs an inverse of size $n - |I_i|$ per cluster, which is what a two-stage method exists to avoid.

4 The gateway model

Definition 1 (Gateway model). *For every cluster j , the residual of the remaining members J_j after linear regression on their own knot is uncorrelated with every asset outside I_j . Equivalently the cross-cluster residual*

$$R_j(p_j) = \Sigma_{J_j,-j} - \Sigma_{J_j p_j} \sigma_{p_j}^{-2} \Sigma_{p_j,-j} \quad (4)$$

is zero.

A member may be correlated with anything, but its correlation with other clusters passes entirely through its knot. The knot is the cluster's only gateway. Each cross-cluster block of Σ then has rank at most one.

This is a strong assumption, and clustering alone does not imply it. It asks that member residuals be uncorrelated both with the other knots and with the other clusters' member residuals.

Proposition 1 (Knots are sufficient). *Under the gateway model, for each cluster i and every u ,*

$$\Sigma_{ii}^c = \Sigma_{ii} - \Sigma_{i,P_{-i}} \Sigma_{P_{-i}P_{-i}}^{-1} \Sigma_{P_{-i},i}, \quad b_i = u_{I_i} - \Sigma_{i,P_{-i}} \Sigma_{P_{-i}P_{-i}}^{-1} u_{P_{-i}}. \quad (5)$$

Proof. The matrix $\Sigma_{i,-i} \Sigma_{-i,-i}^{-1}$ holds the coefficients of the linear regression of cluster i on all other assets. Split the regressors into the other knots P_{-i} and the other remaining members J_{-i} , and write $J_{-i} = \beta P_{-i} + \varepsilon$ for the block-diagonal regression of each set of remaining members on its own knot. The map from (P_{-i}, J_{-i}) to (P_{-i}, ε) is invertible, so the regression may be run on the latter pair. Under the gateway model ε is uncorrelated with cluster i and with P_{-i} , so its coefficients vanish and the coefficients on P_{-i} are $\Sigma_{i,P_{-i}} \Sigma_{P_{-i}P_{-i}}^{-1}$. Substituting into (3) gives (5), with u_{-i} contributing only through its P_{-i} entries. $\square$

Conditioning cluster i on the other knots is the same as conditioning it on everything. The inverse in (5) is $(k-1) \times (k-1)$.

Remark 1 (Relation to Vecchia approximations). *A Vecchia approximation orders the variables and replaces each conditioning set by a small subset of the predecessors, which defines one acyclic approximate joint law [4, 6]. The conditionals are Schur complements. Replacing $-i$ by P_{-i} in (3) is the same truncation of a conditioning set, but it is applied to the full conditional of every cluster at once. Those conditionals are mutually cyclic and no approximate joint law is formed, so the recipe is Vecchia-inspired rather than a Vecchia approximation.*

The gateway model itself does have an exact Vecchia form.

Corollary 1 (Gaussian factorization). *If the returns r are jointly Gaussian, the gateway model holds if and only if*

$$f(r) = f(r_P) \prod_{i=1}^k f(r_{J_i} | r_{P_i}). \quad (6)$$

Proof. Under the model the residuals $\varepsilon_i = r_{J_i} - \beta_i r_{P_i}$ are uncorrelated with r_P and with each other, hence independent under Gaussianity, which gives (6). Conversely (6) makes ε_i independent of every asset outside I_i . $\square$

With the knots ordered first, (6) is a block-Vecchia factorization, a sparser relative of the knot-based cases collected by Katzfuss and Guinness [6]. The covariance results in this note do not need Gaussianity.

Remark 2 (Testing the model). *Regressing each remaining member on its own knot and the other knots, and checking that the other-knot coefficients vanish, does not test the model. Member residuals can be orthogonal to every knot and still be correlated with each other across clusters, and the sufficiency of the knots then fails. The model holds exactly when $R_j(p_j) = 0$ for every j , which can be checked from the sample covariance before any portfolio is built.*

Remark 3 (Measuring the loss and choosing knots). *The raw size of R_j does not measure what is lost when the model fails. A residual with variance δ^2 and covariance $c\delta$ with an outside asset has $R = c\delta \rightarrow 0$, while the Schur correction it carries, c^2 , does not move. The loss for cluster i is the omitted correction*

$$\Delta_i = \Sigma_{i,-i} \Sigma_{-i,-i}^{-1} \Sigma_{-i,i} - \Sigma_{i,P_{-i}} \Sigma_{P_{-i}P_{-i}}^{-1} \Sigma_{P_{-i},i} \succeq 0, \quad (7)$$

the part of cluster i explained by the other assets beyond their knots. A scale-free screen for a candidate knot $p \in I_j$ is the largest canonical correlation between the residual of $I_j \setminus \{p\}$ on p and the assets outside I_j . Both need a large inverse, but they are diagnostics computed once and are not part of the allocation.

Remark 4 (A symmetric factor model). *Suppose instead that every member of cluster j loads on a latent cluster factor, $r_m = \lambda_m f_j + \eta_m$, with idiosyncratic terms uncorrelated with each other and with the factors. Then $R_j(p) = (1 - \rho_p) \lambda_{J_j} \text{Cov}(f_j, r_{-j})$, where $\rho_p = \lambda_p^2 \text{Var}(f_j) / \sigma_p^2$ is the share of the knot's variance explained by the factor. The gateway model holds to the extent the knot is the factor. A knot with a high ρ_p is then the natural choice, and the member most aligned with the cluster's first principal component approximates it.*

5 The bridge

Following [2], damp the conditioning. For $\gamma \in [0, 1]$ define the *conditioned pair* of cluster i ,

$$\begin{aligned} Q_i(\gamma) &= \Sigma_{ii} - \gamma \Sigma_{i, P_{-i}} \Sigma_{P_{-i} P_{-i}}^{-1} \Sigma_{P_{-i}, i}, \\ b_i(\gamma) &= u_{I_i} - \gamma \Sigma_{i, P_{-i}} \Sigma_{P_{-i} P_{-i}}^{-1} u_{P_{-i}}, \end{aligned} \tag{8}$$

and the direction $d_i(\gamma) = Q_i(\gamma)^{-1} b_i(\gamma)$. Let D_γ be the $n \times k$ matrix with $d_i(\gamma)$ in column i , supported on I_i . The bridge portfolio is

$$w_\gamma \propto D_\gamma (D_\gamma^\top \Sigma D_\gamma)^{-1} D_\gamma^\top u. \tag{9}$$

It solves, on the span of the k directions, the problem that $\Sigma^{-1}u$ solves on the whole space. A direction that vanishes is dropped.

Two-tier reading. The right side of (9) does not change when a column of D_γ is rescaled. When every total $\mathbf{1}^\top d_i(\gamma)$ is nonzero, set $v_i = d_i / \mathbf{1}^\top d_i$. Then (9) is the two-stage method (1) with $a \propto (V^\top \Sigma V)^{-1} V^\top u$. The induced outer inputs are the $k \times k$ covariance $V^\top \Sigma V$ and the k -vector $V^\top u$, which is $\mathbf{1}$ when $u = \mathbf{1}$. The unnormalized form (9) has no singularity when a cluster total crosses zero.

Proposition 2 (The two ends of the bridge).

1. At $\gamma = 0$ with $u = \mathbf{1}$ the recipe is NCO with minimum variance at both tiers.
2. Under the gateway model the columns of D_1 are the blocks of $\Sigma^{-1}u$, and $w_1 \propto \Sigma^{-1}u$.

Proof. At $\gamma = 0$ the pair is $(\Sigma_{ii}, \mathbf{1})$, so $d_i = \Sigma_{ii}^{-1} \mathbf{1}$ has a positive total, v_i is the minimum-variance portfolio of the block, and the outer tier is minimum variance across the cluster portfolios. For the second claim, (2) and proposition 1 give $\Sigma^{-1}u = D_1 \mathbf{1}$, which lies in the column span of D_1 . Then $D_1 (D_1^\top \Sigma D_1)^{-1} D_1^\top u = D_1 (D_1^\top \Sigma D_1)^{-1} D_1^\top \Sigma D_1 \mathbf{1} = D_1 \mathbf{1}$. $\square$

γ	Method
0	NCO [1]
(0, 1)	the interior of the bridge
1	unconstrained global minimum variance, under the gateway model

Table 1: The bridge with $u = \mathbf{1}$ and minimum variance at both tiers.

Scope. The right end is the unconstrained, fully invested, long-short solution. No condition on cluster totals is needed for (9). A global solution can hold a self-financing position inside a cluster with zero net weight. The two-tier reading fails there, because no cluster budget times a normalized cluster portfolio represents it, and (9) does not.

Remark 5 (Other objectives). *Proposition 2 is stated for a direction $\Sigma^{-1}u$. With $u = \mu$ the direction $\Sigma^{-1}\mu$ maximizes the Sharpe ratio. Its fully invested multiple does so only when $\mathbf{1}^\top \Sigma^{-1}\mu > 0$. When that total is negative the fully invested multiple has the opposite orientation and the lowest signed Sharpe ratio.*

Mean-variance problems with a budget constraint have solutions that combine $\Sigma^{-1}\mu$ and $\Sigma^{-1}\mathbf{1}$, and each part is covered separately. Risk parity is not of this form. The recipe still runs with a risk-parity outer tier, but nothing is exact at $\gamma = 1$.

Remark 6 (Rules that read only a covariance). *An inner rule that cannot take a constraint vector can be applied after a change of variables. With $y = b_i \circ x$ the constraint $b_i^\top x = 1$ becomes $\mathbf{1}^\top y = 1$ and the covariance becomes $M = \text{diag}(b_i)^{-1} Q_i \text{diag}(b_i)^{-1}$. Apply the rule to M and map back by $x = y/b_i$, coordinate by coordinate. For minimum variance this returns $Q_i^{-1}b_i$. It requires b_i to have no zero entry. Applying minimum variance to M without mapping back returns $\text{diag}(b_i)Q_i^{-1}b_i$, which is a different portfolio.*

Remark 7 (Relation to the tree). *Schur complements compose: the complement of a complement is the complement against the union. So at $\gamma = 1$ the recursion of [2], run down any tree whose leaves are the clusters and stopped there, hands each cluster the pair (3). The flat construction and the iterated one agree at the right end. They differ in the interior, because the tree damps at every split and the damping compounds along the path, while (8) damps each cluster once against all the other knots.*

6 Where to sit on the bridge

The cross-knot regression coefficients in (8) are estimated, and γ is how much of that estimate to trust. The analysis of the tree in [7] transfers in part. This section takes $u = \mathbf{1}$.

Let $\hat{\Sigma}_\tau = \Sigma + \tau E$ be an estimate, where E is a random symmetric matrix whose law is unchanged by $E \mapsto -E$, and $\hat{\Sigma}_\tau \succ 0$ on the range of τ considered. Every quantity in the recipe is computed from $\hat{\Sigma}_\tau$. The partition and the knots are held fixed as τ varies, since reclustering or reselecting knots from the data can destroy smoothness. The portfolio w_γ of (9) is normalized by $\mathbf{1}^\top w_\gamma = 1$, and we assume the normalizing total stays nonzero. Define the expected out-of-sample variance and its noiseless profile,

$$F(\gamma, \tau) = \mathbb{E}[w_\gamma(\hat{\Sigma}_\tau)^\top \Sigma w_\gamma(\hat{\Sigma}_\tau)], \quad V_0(\gamma) = F(\gamma, 0). \quad (10)$$

It is even in τ . We assume F is smooth, put $G(\gamma) = \frac{1}{2} \partial_\tau^2 F(\gamma, 0)$, and assume the expansion $F = V_0 + \tau^2 G + O(\tau^4)$ holds in C^2 as a function of γ .

Proposition 3 (Incremental noise cost). *Let $\Phi(\gamma, \tau) = F(\gamma, \tau) - F(0, \tau) - [V_0(\gamma) - V_0(0)]$. Then $\Phi(\gamma, \tau) = \gamma \tau^2 H(\gamma, \tau)$ for a smooth H .*

Proof. $\Phi(0, \tau) = 0$, so $\Phi = \gamma \Phi_1$ with Φ_1 smooth. Also $\Phi(\gamma, 0) = 0$, so $\Phi_1(\gamma, 0) = 0$, and Φ_1 is even in τ , so $\Phi_1 = \tau^2 H$. $\square$

On the tree, under noise supported only on the top-level cross-block, the hierarchical end does not read that block, and the uncentred cost $F - V_0$ already carries the factor γ [7]. Under whole-matrix noise that fails on the tree as well, because $F(0, \tau)$ is then noisy.

NCO is different even under cross-cluster noise alone. Its outer tier reads cross-cluster covariance at $\gamma = 0$, so NCO has an estimation cost of its own, and the factor γ appears only after centring at the noisy NCO end. The centred statement holds for the general noise of this section, and the cost of moving away from NCO is still of order $\gamma\tau^2$.

Corollary 2 (Separation from NCO). *If $V'_0(0) < 0$ then $\partial_\gamma F(0, \tau) < 0$ for all sufficiently small τ , and NCO is strictly suboptimal.*

Proof. By proposition 3, $\partial_\gamma F(0, \tau) = V'_0(0) + \tau^2 H(0, \tau)$. □

Proposition 4 (The minimum-variance end). *Under the gateway model $V'_0(1) = 0$ and $V''_0(1) = 2\dot{w}^\top \Sigma \dot{w}$, where $\dot{w} = \partial_\gamma w_\gamma$ at $\gamma = 1$. Suppose $\dot{w} \neq 0$. Then for all sufficiently small τ the map $F(\cdot, \tau)$ has a unique stationary point near 1, a strict local minimizer, at*

$$\tilde{\gamma}(\tau) = 1 - \frac{G'(1)}{V''_0(1)} \tau^2 + O(\tau^4). \quad (11)$$

If $G'(1) > 0$ this local minimizer is strictly inside the bridge for small $\tau > 0$. If $G'(1) < 0$ then $\gamma = 1$ is a strict local minimizer of $F(\cdot, \tau)$ on $[0, 1]$. If $G'(1) = 0$ the second order is silent and the fourth or a higher order decides.

Suppose in addition that $\gamma = 1$ is the unique global minimizer of V_0 on $[0, 1]$ and that $F(\cdot, \tau) \rightarrow V_0$ uniformly on $[0, 1]$. Then for all sufficiently small τ these local minimizers are the global optimum over $[0, 1]$.

Proof. Since $\mathbf{1}^\top w_\gamma = 1$ we have $\mathbf{1}^\top \dot{w} = 0$ and $\mathbf{1}^\top \ddot{w} = 0$. By proposition 2, $w_1 = \Sigma^{-1} \mathbf{1} / \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, so $\Sigma w_1 \propto \mathbf{1}$. Hence $V'_0(1) = 2\dot{w}^\top \Sigma w_1 = 0$, and in $V''_0(1) = 2\dot{w}^\top \Sigma \dot{w} + 2\ddot{w}^\top \Sigma w_1$ the second term vanishes. Apply the implicit function theorem to $\partial_\gamma F(\gamma, \tau) = 0$ at $(1, 0)$, where $\partial_\gamma^2 F = V''_0(1) > 0$. This gives (11) and strict convexity of $F(\cdot, \tau)$ on a neighbourhood of 1 that does not depend on τ .

When $G'(1) < 0$ the stationary point lies beyond 1, so $F(\cdot, \tau)$ decreases up to the boundary. Under the additional hypotheses the global minimizers of $F(\cdot, \tau)$ converge to 1, so for small τ they lie in that neighbourhood. □

Gateway exactness makes w_1 the minimum-variance portfolio. It does not by itself exclude another $\gamma < 1$ that produces the same portfolio, which is why uniqueness is a hypothesis of the global statement.

The structure of the bridge does not determine the sign of $G'(1)$. Both signs occur, and each is strict in the examples below, so each persists on a relatively open set within the gateway-model parameterization.

An exact interior optimum. Take clusters $\{1, 2\}$ and $\{3, 4\}$ with knots 1 and 3, and

$$\Sigma = \begin{pmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The estimate replaces the cross-knot entry $\frac{1}{2}$ by $Z \in \{0, 1\}$, each with probability $\frac{1}{2}$. The population and both estimates are positive definite and satisfy the gateway model, because the remaining members are uncorrelated with everything.

On the branch $Z = 0$ the directions do not depend on γ and the population variance is $56/169$. On the branch $Z = 1$ the directions are $d_1 = (1, 1)$ and $d_2 = (x, 1)$ with $x = (1 - \gamma)/(4 - \gamma) \in [0, \frac{1}{4}]$, and the population variance of the resulting portfolio is

$$V(x) = \frac{49x^4 - 7x^3 + 36x^2 - 2x + 6}{2(7x^2 + 3)^2}, \quad V'(x) = \frac{P(x)}{2(7x^2 + 3)^3},$$

with $P(x) = 49x^4 + 84x^3 - 21x^2 + 48x - 6$. On $[0, \frac{1}{4}]$ we have $P'(x) = 196x^3 + 252x^2 - 42x + 48 > 0$, while $P(0) = -6$ and $P(\frac{1}{4}) = 1585/256$. So P has exactly one root $x_\star \approx 0.12823$ there, and

$$\gamma_\star = \frac{1 - 4x_\star}{1 - x_\star} \approx 0.5587$$

is the unique optimum, strictly inside the bridge. In exact arithmetic $F(0) - F(\frac{1}{2}) = 3/1210$ and $F(1) - F(\frac{1}{2}) = 5/1452$.

The same covariance with small symmetric noise, $Z = \frac{1}{2} \pm \tau$, illustrates propositions 3 and 4. There $G'(1) = 31779128936/554523204167 \approx 0.0573$ and $V_0''(1) = 48136/3571279 \approx 0.01348$ exactly, so (11) predicts $\tilde{\gamma}(0.05) \approx 0.9894$ against a computed optimum of 0.9896. The function H of proposition 3 is close to 0.0071 at every $\gamma, \tau \in \{0.05, 0.2\}$.

Full coupling stays locally optimal. Take three clusters, each a knot and one remaining member that is uncorrelated with everything, with member variances 1, 4 and 8. Let the knot covariance and its estimates be

$$S = \begin{pmatrix} 5 & 6 & 5 \\ 6 & \frac{33}{2} & 12 \\ 5 & 12 & \frac{25}{2} \end{pmatrix}, \quad \hat{S}_\tau = S \pm \tau \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

the two signs equally likely. The noise is symmetric, touches only cross-knot entries, and preserves the gateway model. Exact differentiation gives $V_0''(1) \approx 2.3949$ and

$$G'(1) = -\frac{4720238484060245648386156800}{806907939268294475102923640251} \approx -0.0058498 < 0,$$

so by proposition 4 full coupling is a strict local minimizer for all sufficiently small τ .

As a spot check, at $\tau = \frac{1}{10}$ and $\tau = \frac{1}{100}$ exact evaluation gives $F(\gamma, \tau) > F(1, \tau)$ for $\gamma \in \{0.999, 0.99, 0.9, 0.5, 0\}$. This is consistent with a global optimum at full coupling but does not prove one.

Neither end owns the bridge. An interior optimum is not a consequence of the architecture, and a statement that it is generic needs a noise model. The derivatives quoted in both examples are exact, computed by truncated bivariate jets in rational arithmetic.

7 The outer covariance

The gateway model also gives the outer covariance a simple form.

Proposition 5 (Knot covariance plus a ridge). *Under the gateway model, let v_i be any portfolios supported on their clusters. Write $\beta_i = \Sigma_{J_i p_i} / \sigma_{p_i}^2$ for the member betas, $E_i = \Sigma_{J_i J_i} - \beta_i \beta_i^\top \sigma_{p_i}^2$ for the residual covariance, and $\kappa_i = v_{i, p_i} + \beta_i^\top v_{i, J_i}$ for the cluster portfolio's beta to its knot. Then*

$$V^\top \Sigma V = K \Sigma_{PP} K + D, \quad K = \text{diag}(\kappa_i), \quad D = \text{diag}(v_{i, J_i}^\top E_i v_{i, J_i}). \quad (12)$$

Proof. Under the gateway model $\Sigma_{J_i,l} = \beta_i \Sigma_{p_i,l}$ for every asset l outside I_i . Applying this on both sides gives $v_i^\top \Sigma_{I_i I_i} v_l = \kappa_i \kappa_l \Sigma_{p_i p_l}$ for $i \neq l$. On the diagonal, the knot is uncorrelated with the member residual, so $v_i^\top \Sigma_{ii} v_i = \kappa_i^2 \sigma_{p_i}^2 + v_{i,J_i}^\top E_i v_{i,J_i}$. $\square$

The outer problem sees the knot covariance, scaled by the cluster portfolios' knot betas, plus their residual variances on the diagonal. The diagonal term raises the smallest eigenvalue of the outer covariance by at least the smallest residual variance. The remaining members supply a ridge, which is one reading of why the outer tier of NCO is stable.

8 Cost

No step forms an $n \times n$ inverse. The largest linear solve has dimension $\max(k, \max_i |I_i|)$, as in NCO: the outer tier is $k \times k$ and each conditioning inverse is $(k-1) \times (k-1)$. The bridge adds work that NCO does not do: the cross-block products $\Sigma_{i,P_{-i}}$, and a leave-one-knot-out inverse per cluster, obtainable from Σ_{PP}^{-1} by a rank-one downdate.

If the gateway model fails, (5) approximates the full complement in (3), and Δ_i in (7) is what is lost.

9 Closing

The contribution is deliberately small. The identity is block inversion, the truncation of the conditioning set follows Vecchia approximations, and the damping is from [2]. What is added is the model under which one knot per cluster suffices, and the bridge from NCO to the global optimum that it makes cheap.

Proposition 1 may also serve Schur-complementary allocation on the tree, where the complement against a sibling block could be taken against that block's knots. The whole-matrix analysis of [7] rests on a scalar reduction specific to the tree and needs a replacement here. We leave both, and an empirical comparison, to future work.

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